

DSCI 510 – Final Project Progress Report
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Can Qualifying Position Predict Race Results in Recent Formula 1 Seasons?

Project Scope Update

The project scope remains closely aligned with the original proposal. The core research question is still whether qualifying (grid) position can predict race finishing position in Formula 1 under the current ground-effect regulation era. In the current implementation, the analysis uses the 2023–2025 seasons because the OpenF1 API provides historical coverage from 2023 onward. The analytical approach remains the same as proposed: exploratory data analysis combined with a simple linear regression model to evaluate how strongly qualifying position predicts finishing position.

At this stage, the main pipeline has been completed using OpenF1 data. The project successfully loads qualifying and race results, merges them by driver and meeting, computes summary statistics, measures correlation, calculates pole-to-win conversion, summarizes average finishing position by grid slot, and evaluates a linear regression model with MAE, RMSE, and cross-validation. The code also generates chart outputs and summary tables for the final results folder.

Data Sources

OpenF1 API (<https://api.openf1.org/v1>): This REST API is the primary data source used in the current implementation. Python code queries the `/sessions`, `/session_result`, and `/drivers` endpoints. For each Grand Prix, qualifying and race sessions are identified within the same meeting using `session_name` and then matched by `meeting_key`. Session result data is used to extract final classified positions for both qualifying and race sessions, and these are merged into `qualifying_position / race_position` pairs for each driver. The API has been successfully integrated into the project, and tests confirm that data can be retrieved programmatically without authentication.

FastF1 Python Library (<https://docs.fastf1.dev>): This library was part of the original proposal for deeper lap-level and telemetry-based analysis. It is not part of the current submitted implementation, but it remains a possible extension for future work.

Ergast / Kaggle Dataset: This dataset was originally planned as a supplementary or validation source for historical race results. Due to time limits, it is not part of the current submitted implementation. The present analysis focuses on a single API source, OpenF1.

Issues / Difficulties

The main difficulty encountered during implementation was selecting the correct OpenF1 endpoint for final race and qualifying results. Early versions of the code used the `positions` endpoint, which represents time-series position changes during a session rather than clean final classified results. The pipeline was later revised to use the `session_result` endpoint, which better matches the project goal of comparing qualifying outcome with final finishing position.

A second challenge was API reliability across sessions. Some sessions returned missing data or 404 responses, and repeated requests occasionally triggered 429 rate-limit responses. To address this, the project code was updated to handle incomplete sessions more gracefully, retry requests when possible, and continue processing available records instead of failing immediately. As a result, the current pipeline successfully loads a large combined dataset and produces stable analysis outputs.